

to exist in a superposition of multiple classical states simultaneously. This introduces an entirely new layer of complexity and strategy, as players can leverage the principles of superposition and entanglement to create intricate moves and surprise their opponents. The game introduces the concept of quantum moves, where a player can move a piece in a superposition of multiple ways until the move is observed. Additionally, entanglement enables the connection of pieces' states across the board, causing the state of one piece to instantaneously affect another. Quantum Chess challenges players to think beyond the classical constraints of traditional chess, encouraging a deeper understanding of quantum concepts while delivering an intellectually stimulating and entertaining gameplay experience.

Quantum optimisation:

Quantum optimisation represents a revolutionary paradigm in problem-solving, leveraging the computational capabilities of quantum computers to address complex optimisation challenges. Traditional optimisation problems, which arise in fields such as logistics, finance, and artificial intelligence, often become exponentially more challenging as the scale of the problem increases. Quantum optimisation algorithms, like the Quantum Approximate Optimisation Algorithm (QAOA), harness quantum parallelism and interference to explore vast solution spaces efficiently. Quantum optimisation algorithms

excel at finding optimal solutions by leveraging the inherent properties of superposition and entanglement. These algorithms can potentially outperform classical optimisation approaches for certain problem instances, offering a promising avenue for industries seeking to enhance efficiency and streamline decision-making processes.

Quantum computing for cybersecurity and AI

Quantum computing holds immense promise in transforming both cybersecurity and Artificial Intelligence (AI) landscapes. The advent of quantum computers poses a potential threat to classical cryptographic methods, particularly those relying on factorisation and discrete logarithm problems. Conversely, quantum-safe cryptographic algorithms, such as those based on lattice cryptography or hash-based techniques, are being developed to fortify digital security in anticipation of quantum threats. The race to quantum-proof encryption methods is crucial for ensuring the resilience of sensitive data against the exponentially enhanced computational power of quantum adversaries.

On the AI front, quantum computing offers exciting prospects for accelerating machine learning algorithms. Quantum

Machine Learning (QML) algorithms leverage quantum principles to enhance the efficiency of tasks such as pattern recognition, optimisation, and data analysis. Quantum computers, with their ability to process vast data sets and explore complex solution spaces simultaneously, have the potential to outperform classical computers in certain machine learning applications. The synergy between quantum computing, cybersecurity, and AI opens new frontiers for technological advancement, calling for interdisciplinary research to harness quantum capabilities for both securing digital landscapes and enhancing the efficiency of intelligent systems.

It is evident that we stand at the precipice of a transformative era in computational science. The interplay between quantum hardware and software tools, exemplified by platforms like Qiskit, Cirq, and Q# along with cloud services, not only fosters innovation in quantum research but also beckons researchers, developers, and enthusiasts to collectively push the boundaries of our computational capabilities. Quantum technologies hold immense promise for addressing complex problems, from optimisation and cryptography to machine learning and drug discovery. **END** 

 **By: Dr K. Anitha Kumari**

The author is associate professor in the Department of Information Technology, PSG College of Technology, Tamilnadu, India.

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